TRAIN semantic banks

TRAIN noun / verb lemmas

CLIP text encoder *

Noun bank

Verb bank

Semantic aggregation

Semantic aggregation

Fusion stages share structure, not parameters.

Sinusoidal positions

Token prefix

Word embedding

Patch features

CLIP ViT-B/16 Visual encoder *

ROI pooling

Linear projection

768 → 512

Input image I

V

Pn

Object fusion 1

Object fusion 2

T1

Semantic selection

K,V

V1

Self-attention

Q

Cross-attention

Learned gate

Vo

Standard Norm / FFN not expanded

Object Slot MLP

Q

K,V

Masked cross-attention

Output projection

Vo′

Transformer decoder ×3

Vocabulary projection

Softmax

Beam search (K = 5)

Five candidate captions

Image-specific prompts

UPT detector *

Object / relation phrases

CLIP text encoder *

to / tr

Pv

Relation fusion

Vr

Relation Slot MLP

Q

K,V

Masked cross-attention

Output projection

Vr′

YOLO detector *

Boxes B

UPT boxes

UPT–YOLO ROI matching
Class + IoU ≥ 0.5

YOLO boxes B

Slot inputs: text, original Vo, boxes.
Masks: valid linked regions only.

CLIPScore also reads image I.
Object consistency also reads YOLO labels.

Decoder likelihood

CLIPScore *

Object consistency

Weighted selection

Final caption

S = 0.3D + 0.4C + 0.3O

* Frozen

Trainable

Training: cross-entropy → SCST + cross-entropy